Vrushali Chittaranjan

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EDUCATION

International Space University
Masters in Space Studies – Engineering and Space Applications *September 2025 – Present*
Courses: **Satellite Bus**, Rocket Propulsion, **Flight Dynamics**, Remote Sensing, MBSE, **Space Robotics**
Ramaiah Institute of Technology *July 2021*
Bachelors of Engineering in Electronics and Instrumentation (Major GPA= 8.015/10)
Sophia High School (10+2) *July 2017*
Matriculation: - ISC Science

TECHNICAL SKILLS

Programming & Scripting Languages	***Data Science, AI & Analytics***
General Purpose: C, C++, Java, JavaScript, Python, Julia, R	Machine Learning/AI: PyTorch, Scikit-learn, Keras
Embedded & Hardware Description: Embedded C++, ARM Processor Coding (using C), VHDL, Verilog, Arduino	Data Visualization & BI: Tableau, SAP BO (Business Objects)
Database: SQL, PLSQL	***Statistical Computing:*** *R & R Studio*
Engineering, Design & Simulation	***Automation & Software Testing***
CAD & Modeling: AutoCAD, SolidWorks	Automation: UiPath Automation, Selenium (via Eclipse)
Systems & Control: Simulink, MBSE (Model-Based Systems Engineering), PLC & SCADA	Version Control: Git
Hardware & Electronics: FPGA modules, Arduino	
	Development Platforms & Other
Space & Specialized Engineering	Mobile: Android
Astrodynamics & Simulation: STK (Satellite Tool Kit), RocketPy, Topcat	IDE/Environment: Eclipse, R Studio
Remote Sensing & GIS: Q-GIS, Google Earth	Corporate Platforms: Oracle Analytics Cloud, Fusion, APEX, SQL, OraDB
Quantum Computing: IBM Qiskit	

PROJECT AND RESEARCH EXPERIENCE

International Space University
Flight Dynamics Lead and Rocket Design for Mars Rover and Drone Project ***September 2025 – Present***
- Developed an UI for the mission which acts as a center interface to monitor all the workings of drone and rover
- In the process of designing an ultra-light weight Copter/Drone with multispectral cameras
- Simulation of **Blade Design Optimizations** for Martian atmosphere with Low Reynold's number and high Mach at the blade tip
- Integration of **Machine Learning Models** with a few layers to detect and map terrains as well as help model automated drone paths. In the process of Integrating these modules for autonomous flight path and real-time decision-making capabilities using Python, **Pytorch** and **Keratese**

Research Project – Enhancement and Conceptual design of Air Breathing Rocket Engine ***September 2025 – Present***
- Preliminary research on fusion of air breathing rocket
- Currently working on Blade Design Optimization for vertical lift off before transition into rocket mode and building a tool to visualize the **CFD** patterns for this scenario
- Using python libraries such **RocketPy** to simulate our rocket path and optimize our design and requirements.
- Aiming to prove fuel efficiency and reduce launch costs to a great extent especially for sub orbital commercial flight

Project Manager – CubeSat for Monitoring Deforestation with Obstacle Avoidance Capabilities ***October 2025***
- Designed the structure for a **1U Cube Satellite** with Payload being a Camera for monitoring Deforestation
- Designed the ADCS system integrated with Obstacle Detection sensors to prevent collisions as well as Sun Sensor to simulate SUN and ECLIPSE cycles that would help the cubeSat detect charge and discharge cycles using Solar Panels
- Developed a robust EPS circuit system for to power all subsystems without damaging the parts and extracting maximum output.
- Integrated the COMMS subsystem to make sure the telemetry received was absolute, precise and accurate.

WORK EXPERIENCE

Oracle Cloud Engineer | Data Integration & Analytics Engineer ***August 2021- August 2024***
- System Architecture: Architected and validated complex data objects by integrating heterogeneous sources (SAP, Tableau) into Oracle Analytics Cloud (OAC) across DEV, TEST, and PROD environments.
- Enterprise Integration: Engineered end-to-end inbound/outbound integrations using Oracle Integration Cloud (OIC) and REST API adapters to synchronize data between Cloud and On-Premise systems.
- AI & Optimization: Implemented AI-driven Intelligent Document Recognition (IDR), increasing operational efficiency by 67% for high-volume crisis response (UNHCR).
- Automation Engineering: Developed UI Path and Selenium-based automation frameworks for large-scale migrations (20,000+ records) with 96% accuracy; logic adopted as Oracle Intellectual Property.
- Full-Stack Development: Designed custom financial models using PL/SQL and developed low-code applications via Oracle APEX for mission-critical reporting.
- Certifications: OCI Enterprise Analytics, AI Foundation, and Oracle Analytics Cloud.

RESEARCH AND TECHNICAL EXPERIENCE

International Astronautical Conference – ***Undergraduate Researcher*** ***April 19th 2022 – September 22nd 2022***

- Proposed a novel method for unfolding of cancerous proteins using **Magnetic Nanoparticles (MNP's)** inspace under microgravity conditions to study its structure.
- Collated and surveyed the required research and evidence from a medical and engineering perspective to claim such anexperiment is plausible in microgravity.
- Evaluated and executed the Random Movement Algorithm used to untangle proteins on a simulation tool to help analyze the degree of unfolding. Defined the required thermodynamic parameters to maintain the stabilityof the protein.
- Presented this paper at International Astronautical Congress IAC 2022 Paris

PROJECT EXPERIENCE

Indian Space Research Organization

Project Head - Research Intern ***April 1st 2021 – July 13st 2021***

- Engineered a code for the **Synchronization of three Clocks** for a remote satellite that cannot be controlled by the ground station.
- It was implemented with **Triple Modular Redundancy** to prevent failure of other instruments in the satellite.
- Accuracy of 99% Synchronization was achieved was correct to a 0.3 nanoseconds which was a vital factor for the execution of other modules that depend on synchronization in the satellite.
- Programmed and executes this entire code on Model Sim using **VHDL Language**
- Received Excellent Project Award from ISRO

Ramaiah Institute of Technology (part of undergraduate credits)

Project Coordinator ***January 12th 2021 – July 31st 2021***

- Final Year Project from the months of January to July evaluated for 15 credits as per marks sheet.
- Developed novel machine for remote physiotherapy with enhanced adaptive Machine Learning for dynamic learning and results. It was a solution-based project aiding people with arthritis and people with knee-injuries of all age groups
- Implemented K-means clustering algorithm to segregate data into different groups and 1-D Convolutional Neural Network to train the Machine Learning model using **Pytorch** and **Keratese**
- Executed this Machine Learning model which derived a KQM (Knee Quality Metric) determining the severity of the knee and customizing the physiotherapy for that severity.
- Created a custom-made mobile app using Android Language
- This project was nominated by the Department of Electronics and Instrumentation to apply for a patent.

Indian Institute of Science – ***Research Intern***

Quantum Cryptography ***July 15th 2020– December 10th 2020***

- Researched core concepts of **Quantum Cryptography** such as **Quantum Key Distribution** and Quantum entanglement and surveyed the technical aspects regarding quantum security
- Implemented **Bernstein-Vazirani Algorithm** with the help of various quantum gates (such as Hadamard transformation, Toffoli gate, etc.) on Qis kit
- Analyzed Quantum Teleportation algorithm by building a Quantum Teleportation circuit on Qis kit.
- Studied and implemented **encryption** and decryption methods for **Quantum One Time Pad**
- Investigated the **BB84 protocol** used for quantum communication and how it can be used as a secure key distribution protocol.

Quantum Entanglement ***July 13th 2019 – December 7th 2019***

- Initial research included studying Feynman lectures to understand the basics of Quantum Mechanics
- Focused my core research on Quantum Entanglement of Superconducting Qubits where I explored on specialized topics such as quantum distillation, distortions of the photon and entropy of a quantum system.

PUBLICATIONS

- Sridhar Kashyap, Vasuki Venkatesh, M.K. Pushpa, Sidharth Narasimhan, **Vrushali Chittaranjan**, Adaptive, AI-based automated knee physiotherapy system, *Global Transitions Proceedings*, Volume 2, Issue 2, 2021, Pages 484-491, ISSN2666-285X
- **Vrushali Chittaranjan**, Rubiya Shaikh Protein Unfolding – Redefining the Future of Proteins in Space and As a Cure for Various Cancerous Diseases, *International Astronautical Congress*, Paper number, IAC-22, A2,7,9, x70395

HONOURS AND AWARDS

- Second Place for MSS Space Robotics Competition at ISU
- Best Project Award for year 2021 – Automated Knee Physiotherapy Machine from Ramaiah Institute of Technology
- Excellent Performance for Project – TMR System for Clock Synchronization from Indian Space Research Organization – **ISRO** for the year 2021

LANGUAGES

- English – Fluent
- Hindi – Fluent
- French – A1 and still learning
- German – A1 and still learning